

Topology-Aware Differentiable Triangle-Soup Reconstruction via Persistent Homology

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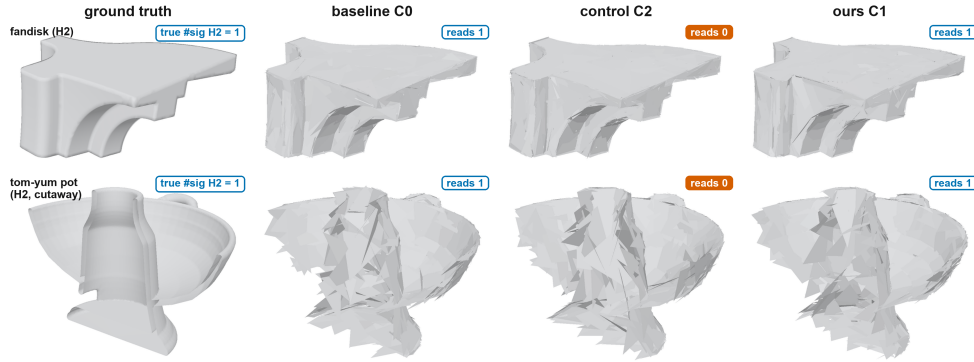


Figure 1. Geometry alone cannot distinguish topology; measurement can. Final reconstructions (seed 0), identical budgets: ground truth; baseline (C0); the norm-matched *non-topological* repulsion control (C2); the persistence loss (C1, ours) — equal-or-better Chamfer on every row (§5.2). A topological error need not be visible, so each cell is badged with its *measured* significant-feature count (blue outline = correct, filled orange = wrong; identical across seeds): the control erases both voids — and collapses a loop on bob (suppl. Fig. S8) — while the loss reads correct everywhere, cutting tail error up to $10.4\times$ (group means $1.5\times/4.9\times$; Tables 1–2).

Abstract

Differentiable triangle-soup reconstruction inherits a limitation from its objective: photometric and geometric losses cannot measure topology, so a reconstruction with a collapsed loop or a punctured enclosed void can score exactly as well as a correct one (on Chamfer-equal probes the diagrams differ 35–40 $\times$ in bottleneck distance). The standard implicit remedy — steer where the resampler spends its budget — does not repair this: in a controlled allocation study, a topology-informed prior is largely matched by an equally wide random one, and no prior shape repairs loops. We therefore move topology into the objective: a differentiable persistence term compares the evolving surface’s diagram, measured on live surface samples, to a fixed target; gradients flow through a pair-frozen backward re-expressing matched birth/death simplices as closed-form circumradii (exact for Gabriel simplices, verified per pairing), plus a recruitment term restoring the gradient optimal matching provably lacks when a target feature is missing; one ratio knob calibrates the loss against the photo-

metric gradient, no curriculum needed. Every claim passes a channel-controlled verdict: the loss must beat a norm-matched non-topological control through the identical gradient channel, at Chamfer parity. Under that rule the loss is topology-specific for enclosed voids (4.0–7.9 $\times$ lower error) and — the class every allocation prior failed — for loops (2.3 $\times$, zero phantom handles, while the control collapses one); loss and prior compose; component counts (H0) are a null result, matched by the generic control. The verdicts replicate without per-shape tuning on eight external genus-known meshes in two observable-class groups (pre-registered group means: loops 1.52 $\times$, voids 4.87 $\times$; one a raw artist mesh, ingested and certified; the one non-pass is a no-headroom null, its already-correct baseline left unharmed), degrading gracefully under sensor noise. Scope: all evidence is synthetic and single-machine — closed surfaces plus a first open-surface probe — with the target diagram known in advance, the regime where correction can be measured exactly; real scans and blind capture are future work. Prescription: correct topology in the loss, allocate

wide, combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [91], periodically resampling under a fixed budget. Photometric losses are topologically blind: reconstructions differing in genus, connectivity, or enclosed voids can be indistinguishable to Chamfer and Hausdorff while persistence diagrams separate them $35\text{--}40\times$ in bottleneck distance on Chamfer-equal probes (§2; suppl. Table S1). Fig. 1 previews the consequence: identical budgets, equal-or-better Chamfer, topology standing or falling with the objective.

A decade of connectivity-free reconstruction — point splats, Gaussians, and lately a return to triangle soups for their native graphics-pipeline fit [11, 40, 91] — has treated topology errors as symptoms of geometric under-provision: represent the surface better, or allocate the budget more wisely, and topology should follow. We tested the allocation half first (suppl. §A). It fails: a topological prior helps enclosed voids, but any equally-wide *non-topological* prior reproduces most of the gain, and no prior shape repairs loops — concentrating manufactures phantom handles, spreading never beats a random control.

The null result relocates the root cause: not *where* triangles are spent, but *what the objective measures* — a photometric loss is a local, pixel-space signal; a loop or an enclosed void is a global, categorical property of the surface. This paper therefore moves topology into the *objective*: a differentiable persistence term whose gradients act directly on features’ birth/death values rather than on where triangles respawn.

Three research hypotheses structure the study, each tested by a pre-designed condition:

1. **RH1 (diagnosis)**: topology errors originate in the training objective, not in primitive allocation — tested by the allocation study above.
2. **RH2 (specificity)**: a persistence term improves topology in ways no norm-matched non-topological regularizer through the identical channel can reproduce — the control C2 (§3.5).
3. **RH3 (complementarity)**: objective and allocation are complementary channels, not substitutes — the composition C5.

Concretely, we contribute **a matched persistence loss with a pair-frozen backward** (§3): exact persistence forward (GUDHI alpha complex), backward through closed-form circumradii — Gabriel-exact for 100% of significant pairs in practice; **a recruitment term for missing features**: optimal matching yields zero gradient when a target feature has no live counterpart; recruitment attaches the nearest live bar, and removing it collapses the loop-class win to baseline

(§5); **one calibrated knob**: λ set by a gradient-norm ratio ρ , flat over a $10\times$ range, no curriculum needed — an ablation, not an assumption; and **channel-controlled evaluation**: every claim must beat a norm-matched *non-topological* control through the identical channel (§3.5).

The loss proves topology-specific on the class no prior ever won — loops, zero phantom handles — confirming RH2; the channels compose (RH3); the H0 class is matched by the generic control, sharpening RH1: topological guidance earns its keep where geometry and topology decouple.

2. Related work

A decade of gradient-based reconstruction: four positions on connectivity. OpenDR [65], neural mesh rendering [54], SoftRas [64], DIB-R [17] and nvdiffrast [60] established gradient flow from pixels to geometry; reconstruction work since splits on what that geometry should be. (i) *Implicit fields* — learned occupancy and signed distance [68, 76], volume-rendered from NeRF [69] through NeuS [94] to Neuralangelo [63] — are watertight and topologically consistent *by construction* (the strongest counterposition), at field-evaluation and extraction cost. (ii) *Fixed-connectivity templates* deformed per input [38, 93] keep manifoldness but freeze topology at the template’s genus. (iii) *Differentiable iso-surfacing* re-extracts the mesh from a volumetric scaffold at every step [72, 86, 87] (Shape-as-Points bridges point soups to watertight implicits [78]) — topological freedom at grid cost. (iv) *Connectivity-free primitive soups* — point splatting [101], patch atlases [34], Gaussians [58] and their surface-accurate variants [35, 47], and a 2025–26 return to native triangles (splatted [40], soft-connected [11], rasterized by DiffSoup [91]) — are the fastest and the easiest to budget, but even their surface-minded members regularize *geometry* (alignment, flatness, connectivity forces), never a measured topology. Mesh-Splatting, conversely, renders *already-connected* opaque meshes differentially [39]: connectivity as input; ours a free soup with measured topology. (i) and (iii) solve topology representationally; (iv) had not addressed it. Cut by *optimization philosophy* rather than representation, the decade holds *geometry-oriented* objectives (every method above), *sampling-oriented* steering of where a fixed objective spends primitives (below), and an empty third family — *topology-oriented* objectives that measure the property and differentiate through it (suppl. Table S6). Prior work occupies every pairwise intersection of the section’s axes — explicit triangles, differentiable optimization, topology in the objective; this paper takes the three-way one, at the soup’s budget and speed. We build on DiffSoup, whose in-loop prune/respawn resampler is the very mechanism our allocation study (suppl. §A) steered; here the resampler stays untouched and only the objective changes. Vertex-domain preconditioning [73] is complementary: it changes *how* ver-

tices move, we change *why*.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper’s problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal *by construction* (0.916/0.628/0.977% per pair), the discriminating-dimension bottleneck separates correct from broken topology by $35\text{--}40\times$ — thin handle ≈ 0 vs .0302, bridged merge .0014 vs .0574, punctured void .0040 vs .1385 — and Hausdorff₉₅ even *prefers* the wrong candidate in two of three cases (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck 33–53% below baseline — but a width-matched *random* control recovers most of that gain, leaving a small, shape-dependent residual: the prior’s value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the *non-topological* wide control (.0267 vs the spread prior’s .0316), and the concentrated field manufactures phantom handles (4.4 significant loops vs the true 2); C5 below reuses the study’s strongest field (B4; suppl. §A).

Persistence in learning objectives. Persistent homology [25] with its stability guarantees [21] has been used as a trainable signal via differentiable sorting of filtration values [9], for topology-preserving image segmentation [20, 45], point-cloud and shape optimization and 3D volume prediction [8, 79, 92], and diagram-metric optimization with convergence analysis [13]; barcode-valued maps have since been given a general differential framework [62], and the per-step cost of persistence-guided descent attacked directly [74] — yet never as the objective of an *explicit triangle-based* reconstruction. Nearest our setting, concurrent work brings persistence to reconstruction at other, complementary entry points: an image-space barcode term for Gaussian splatting [88], homology-guided camera placement [30] or a genus-matched template [29] in mesh inverse rendering, and a connected-component constraint on implicits fitted to point clouds [51]. Partition our setting’s three requirements — (i) a topological quantity of the evolving *surface* measured differentially in the objective, (ii) image-based training through a renderer, (iii) a fixed primitive budget — and each neighbour satisfies at most one (suppl. §D.7): the image-space barcode term never measures the surface (i); the point-cloud pipeline forgoes renderer and budget (ii, iii); the template line *assumes* the genus we measure (i). A head-to-head benchmark is therefore ill-posed (inputs, outputs, and cost regimes differ); our evaluation instead pits the loss against norm-matched controls through its own channel (§4). The nearest soup-side neighbour regularizes coherence with *soft connectivity forces* [11] — local continuity pressure, no measured topology; we replace that heuristic with the mea-

surement. Our backward pass is in the persistence family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex’s birth/death as a closed-form circumradius — with three pipeline-specific ingredients. (i) *Alpha-complex pairing over $\alpha\times\text{area}$ surface samples* of a live triangle soup — not image grids or raw clouds — couples the diagram to the geometry actually being trained. (ii) A *Gabriel-exactness gate*: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — certifying the frozen gradient rather than assuming it. (iii) *Recruitment for missing features*: optimal partial matching has zero gradient exactly when a target feature is entirely absent, so we trade metric fidelity for a restored descent direction (§3), quantifying both its pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [4, 52, 85], and budgeted remeshing has a long lineage [3, 7, 23, 31, 43]; our contribution is orthogonal — the budget mechanics stay DiffSoup’s, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8 L_1 + 0.2 \text{SSIM}$) [91], which we adopt *unchanged*: every ablation below isolates the topology channel, never photometric tuning. We add

$$L = L_{\text{photo}} + \lambda(t) L_{\text{topo}}(V), \quad (1)$$

where V are the soup’s vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed *target bundle* $\mathcal{T}$: the significant finite bars of the ground-truth shape’s diagram, computed once at the density M the loss will see (alpha values shift with spacing: density matching is a correctness requirement), diagonal-normalized, and locked for all live evaluations. Throughout, a bar is *significant* iff its lifetime exceeds the measurement floor $6 r_{\text{med}}(M)$ (r_{med} = median nearest-neighbour sample spacing; derived and validated in §6).

3.2. Live diagram and pair-frozen backward

At each refresh we draw M surface samples ($M=2048$ unless the measurement-floor rule of §4 dictates a denser bundle) $X_j = \sum_i w_{ij} V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha)\times\text{area}$ -weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach

vertices while the combinatorics stay fixed. The alpha complex of X [24], computed exactly with GUDHI [67], yields persistence pairs (birth simplex, death simplex) per dimension (suppl. Fig. S9). For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar’s coordinates are closed-form functions of a few sample positions, all exactly differentiable:

$$R_{\text{edge}} = \frac{1}{2} \|p_1 - p_0\|, \quad R_{\text{tri}} = \frac{\|u\| \|v\| \|u-v\|}{2 \|u \times v\|}, \quad (2)$$

$$R_{\text{tet}} = \|y\| \quad \text{with} \quad Ay = \frac{1}{2} \text{diag}(AA^\top),$$

where $p_0, \dots, p_k$ are the k -simplex’s sample positions, $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the rows $p_i - p_0$ (the circumcenter is $p_0 + y$). Edges carry H0 deaths and H1 births, triangles H1 deaths and H2 births, tetrahedra H2 deaths. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); a pair failing this Gabriel test is *detached* — it keeps its (correct) value in the loss but propagates no gradient, so an exactness failure can never inject a wrong one. In practice the gate never fires: 100% of significant pairs are Gabriel-exact on all six shapes’ clean clouds, and the per-refresh failure count is zero at every one of the 5,500+ logged refreshes (all 2,400 noisy ones included, §5) — a constant pass rate, not an aggregate. Stronger fallbacks (dropping pairs, subgradients, locally raising M) remain available. The pairing is treated as locally constant — the standard pair-frozen device [9, 13] — refreshed every $K=10$ steps and, necessarily, after every resampling event (which replaces the primitive tensors wholesale); $K=10$ is an amortization choice, not a tuned knob — it bounds refresh cost (174 ms each; the fixed ≈ 46 s of §5) while the pairing stays local.

3.3. Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j) \text{ matched}} [(b_i - b_j^*)^2 + (d_i - d_j^*)^2] + \sum_{i \text{ unmatched}} \left(\frac{d_i - b_i}{2} \right)^2 + L_{\text{recruit}}. \quad (3)$$

The first term pulls matched features to their targets; the second pushes spurious features to the diagonal; the third exists because optimal matching has a provable failure mode:

Lemma 1 *If a target bar $t \in \mathcal{T}$ is unmatched under the optimal partial matching, then L_{topo} without L_{recruit} is locally independent of t : no component of its gradient in V acts to create the missing feature.*

Proof sketch. The unmatched target’s augmented cost is its diagonal penalty $((d_t^* - b_t^*)/2)^2$, constant in the live coordinates; every V -dependent term of Eq. 3 references live bars only, and the matching is locally constant in V (the pair-frozen regime) — no gradient term is directed at t . $\square$

Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target bars, processed in decreasing persistence order, and let the candidate pool $\mathcal{A}$ hold every live bar not claimed by the matching — unmatched significant bars and, crucially, *sub-threshold* bars, which matching never sees. Each missing target greedily claims its nearest pool bar, without replacement, under the same squared birth-death metric as the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} [(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2], \quad (4)$$

$$k(j) = \arg \min_{k \in \mathcal{A}_j} [(b_k - b_j^*)^2 + (d_k - d_j^*)^2],$$

where $\mathcal{A}_j$ is the pool remaining after earlier recruitments; recruited bars are excluded from the diagonal term and back-propagate through the same circumradius machinery. This deliberately departs from metric-faithful W_2 ; §5 probes its location-blindness pathology directly, and C6 shows the term is load-bearing. Greedy rather than optimal assignment is deliberate: in every logged regime the missing-target set is a single bar (torus: one unreached target at every refresh, every seed; §5), where the two coincide; when they diverge, decreasing-persistence order spends the pool on the most significant features first.

3.4. Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless ($\rho=0.1$ throughout); the tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, a constant- λ ablation matches the ramped version, and alternative ramp windows land within seed noise (suppl. Table S8) — calibration, not scheduling, makes the knob safe.

3.5. Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: $+$ a short-range repulsion on the *same* frozen samples, calibrated with the *same* ρ — the norm-matched control, the RH2 test (C2g: a gentler radius). C3: C1 without the ramp. (C4, curvature-weighted, is deferred; the numbering gap keeps condition tags comparable across studies.) C5: C1 $+$ the best prior of our allocation study (the spread topological field B4; suppl. §A) — the RH3 test. C6: C1 without recruitment (pure matching $+$ diagonal) — does the term matter in-loop, or only for cold-start repair?

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout $\Rightarrow$ 21–63 training views): sphere and cube (void class H2, $\beta=(1, 0, 1)$, budget $N=1200$), torus (loop class H1, $(1, 2, 1)$, $N=700$), two spheres (component class H0, $(2, 0, 2)$, $N=700$), and genus-2 double torus as a stretch case $((1, 4, 1)$, $N=2000$). Budgets are the allocation study’s headroom points (suppl. §A) — tight enough that baseline topology is imperfect. Eight external, genus-known meshes probe generality beyond the analytic family, organized as the study’s two observable classes (§5.2): a *loop group* — *bob* (genus 1), *rocker-arm* (1), the CGAL *eight* (2) — and a *void group* — *spot*, *fandisk*, *armadillo*, *horse* (all genus 0), the *tom-yum pot* (3, below); sources, licenses and certificates: suppl. §G. Seven pass the mesh certificate below as-is; five further candidates failed it (open boundaries, non-manifold junctions) and were excluded, never repaired. Membership, observable class and density were fixed by the pre-registered staircase rule *before* any training (rocker-arm: torus precedent, H1-restricted at $M=2048$; eight: the pot’s, full $(1, 4, 1)$ bundle at $M=8192$). Scenes: the same pipeline (48 views, 200 px), unchanged class budgets (trio pilot: tails .043–.054, in the analytic band). The fourth, the *tom-yum pot* (Fig. 1, bottom row; aluminium showcase: suppl. Fig. S7), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup (9 overlapping open shells, 232 non-manifold junction edges) with no trustworthy ground-truth topology — the same defect that excluded ShapeNet. An ingest-and-certify pass (exact weld, offset-solidification into a thin closed shell, then *certification by measurement*: exact simplicial homology cross-checked against per-component Euler characteristics, both independent of the loss’s sampled alpha complex; watertightness; bit-identical rebuilds) yields one body of genus 3, $\beta=(1, 6, 1)$ exact, rendered and budgeted as a void-class shape ($N=1200$). This ingest pipeline is a test fixture here, not a claimed contribution. Where the other externals arrive clean enough to certify as-is, the pot’s topology exists only *after* ingest — a controlled real-world proxy, not a blind test: what real scans will need (mechanics and challenge inventory: suppl. §B, Table S4). The pot also exercises the floor rule of §6 prospectively: every H1 loop of the certified shell sits below the $6 r_{\text{med}}$ floor at every working M , while the narrow flue mouth caps at small α — an H2 void on the pot axis, first clearing the floor at $M=8192$ (margin 1.20–1.23 $\times$, five sampling seeds). The pre-registered rule builds the bundle there (a floor-rule density deviation later repeated by eight; the loss samples at the same density): exactly one significant bar, stable to re-sampling ($\sim 10^{-5}$).

Protocol. All conditions share one implementation: C0 is the unmodified DiffSoup trainer; the loss enters as one additive term, and disabled it executes none of the added code (no renderer or resampler edits). 2,500 steps; resampling every 100 steps (untouched); $\rho=0.1$, ramp 0.2:0.5, refresh $K=10$, $M=2048$; decisive shapes (sphere, torus) and the double torus run 5 seeds (the former with the full condition set incl. C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2). (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss *plan* sees with i.i.d. Gaussian noise at every refresh: the plan decides on the noisy observation, the pulls apply to the true live positions. The torus restricts the loss to H1 (its own void bar is sub-threshold at $M=2048$ — a loss must not act on features it cannot reliably measure); the double torus likewise targets its two measurable loops of four. The generality wave runs C0/C1/C2, three seeds, each shape at its staircase-fixed bundle (bob: full loops+void; rocker-arm: H1 only; eight and the pot: $M=8192$).

Metrics and verdict. Primary: bottleneck to the target diagram in the discriminating dimension, tail-averaged (last three dumps, steps 2,300–2,500), seed-averaged (20k eval samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Reporting policy: mean $\pm$ sd over seeds; Welch σ only where both arms ran five seeds; three-seed verdicts rest on disjoint per-seed ranges (suppl. Table S7). *Verdict rule*: C1 must beat C0 **and** every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not one. *Chamfer parity*, formally: the passing arm’s seed-mean Chamfer must satisfy $\text{Ch}(C1) \leq 1.15 \text{Ch}(C0)$; the 15% headroom absorbs seed variance and nondeterminism (§7) and never did any work — every passing arm’s Chamfer was at or below baseline.

5. Results

A gate first: the loss alone (no photometric term, Adam on points) repairs all four controlled probes — a punctured void closes (8.8 $\times$), a spurious handle dies at the diagonal, mis-spaced components separate (1229 $\times$), a bridged merge with *no* significant live H0 bar (Lemma 1’s case) is severed (450 $\times$, β_0 1 $\rightarrow$ 2); the predicted location-blindness did not materialize (suppl. §C, Fig. S5). Raw-cloud sanity toys; multipliers not comparable to training.

5.1. Controlled comparison: analytic shapes

Table 1 (the *C-matrix*: every condition against every analytic shape) reports tail bottleneck (mean $\pm$ sd over seeds) in each shape’s discriminating dimension; Chamfer in the text where it matters; the full training trajectories for the two decisive shapes are in suppl. Fig. S6 (line = seed mean, band = seed min–max).

Table 1. Tail bottleneck to target (mean $\pm$ sd; lower is better); $\times$ = vs C0; verdict: C1 < C0 and < every control at Chamfer parity. Seeds: 5 (sphere, torus, double torus — saturated identically), 3 (cube, two spheres). C6 = no recruitment.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior	C6 no-recruit
sphere (H2)	.0623 $\pm$.0063	.0156$\pm$.0013 (4.0 $\times$)	.1372 (worse)	.0151 $\pm$.0004	.0082$\pm$.0007 (7.6 $\times$)	.0166 $\pm$.0005
cube (H2)	.0582 $\pm$.0046	.0074$\pm$.0011 (7.9 $\times$)	.1346 (β_2 1 $\rightarrow$ 0)	—	—	—
torus (H1)	.0424 $\pm$.0031	.0180$\pm$.0016 (2.3 $\times$)	.0457 (β_1 2 $\rightarrow$ 1)	.0155 $\pm$.0015	.0133$\pm$.0004 (3.2 $\times$)	.0411 $\pm$.0038
two spheres (H0)	.0065 $\pm$.0008	.0007 $\pm$.0002 (9.5 $\times$)	.0008 (ties C1)	—	—	—
double torus (H1)	.0264 $\pm$.0000	.0264 $\pm$.0000 (saturated)	.0264 $\pm$.0000	—	—	—

Voids (H2): topology-specific, large. Sphere 4.0 $\times$ (16.2 σ Welch vs C0, five seeds), cube 7.9 $\times$ (three seeds: ranges disjoint, C1 .0064–.0086 vs C0 .0551–.0636), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is *destructive*, not merely weaker: it pins the void’s death at its repulsion equilibrium (2.2 $\times$ worse, seed-identical) and erases the cube’s void outright; no tested strength reproduces any gain — C1 < C0 < C2g < C2 (§7; spread: suppl. Fig. S3; mechanism: §6).

Loops (H1): the channel’s distinguishing result. No prior shape won this class (suppl. §A). The loss cuts torus H1 error 2.3 $\times$ with *zero phantom handles in all five seeds* (count trajectories flat at the true $\beta_1=2$, suppl. Fig. S4), while the control collapses one loop by step $\sim$ 800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not (mechanism: §6).

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error 9.5 $\times$, but the generic control matches it (.0008 vs .0007); the increment appears only in Chamfer (0.54 vs 1.94). Three channels agree: β_0 is carried by shell count and separation.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 (7.6 $\times$ vs C0, $\approx$ 2 $\times$ vs C1 alone, $\approx$ 4.4 $\times$ below the prior alone), torus .0133 — with the best sphere Chamfer (torus .502, trailing only C3’s .495). Allocation and value-tuning are complementary, not substitutes.

Stretch case (genus 2): saturated, honestly. Every double-torus arm returns exactly .0264: the two tube loops are unexpressable at feasible budgets — the *representation’s* ceiling, not the channel’s (suppl. §B; floors: §6).

Ablations. The ρ -response is flat over a 10 $\times$ range (suppl. Table S8); C3 (no curriculum) $\approx$ C1 on both decisive shapes and off the analytic family (bob: C3 .0171 $\pm$.0026 vs C1

.0208 $\pm$.0008). Overhead, honestly: the loss adds a *fixed* $\approx$ 46 s per 2,500-step run (refresh 174 ms at $M=2048$ every 10 steps) — more than the 33–41 s photometric baseline on these deliberately tiny scenes, but the cost scales $O(M \log M)$ per refresh, not with scene or image complexity, so its share of a run whose photometric cost is T is $46/(T+46)$ — an arithmetic projection, $\approx$ 7% of a ten-minute run — independent of image count and resolution; no parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal) collapses the torus win entirely: .0411 $\pm$.0038, statistically baseline (0.6 σ from C0, 12.4 σ worse than C1) at baseline Chamfer, though both loops still *exist* (#sig H1 = 2 every seed; suppl. Fig. S6). The logs locate the mechanism: one torus loop is significant at every refresh, the other chronically *below* threshold (all five seeds, all 200 refreshes) — recruitment is the second loop’s sole gradient path; on this class it *is* the mechanism, not a cold-start device. On the sphere it never fires (0 of 200, all seeds): C6 is loss-identical to C1, and the 1.5 σ tail gap (.0166 vs .0156) measures pure run-to-run noise (§7).

Sensor-noise stress (C7/C7h). With the plan’s cloud perturbed at every refresh (σ = 0.5%/1% of the diagonal $\approx$ 0.5/1.0 spacings, §4), the loss degrades gradually and never harms: torus .0210 $\pm$.0018 / .0259 $\pm$.0022 (2.0 $\times$ /1.6 $\times$ below baseline, vs C1’s 2.3 $\times$), sphere .0235 $\pm$.0009 / .0328 $\pm$.0013 (2.7 $\times$ /1.9 $\times$, vs 4.0 $\times$), at equal-or-better Chamfer, correct counts in every run (suppl. Fig. S6), zero Gabriel failures in all 2,400 noisy refreshes (internals: suppl. §B).

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule — on the eight external genus-known meshes of §4’s two groups, with no per-shape tuning. Seven of eight **pass** the verdict rule at equal-or-better Chamfer (Table 2; original-wave trajectories in suppl. Fig. S2); the eighth, rocker-arm, is a *no-headroom null*, not a counterexample (obs. 5). The control

Table 2. Generality wave, two observable-class groups (mean $\pm$ sd, 3 seeds; $\times$ = vs C0; *group mean* $\pm$ sd over member shapes, pre-registered). Seven of eight **pass** Table 1’s verdict rule; rocker-arm is a no-headroom null (counts intact in every arm). *seed-exact unmatched-target pins (§6). Per-seed values: Table S7; prove-nance, certificates, staircases: suppl. §G.

shape	C0	C1 loss	C2 control
bob	.0426 $\pm$.0012	.0208$\pm$.0008 (2.1 $\times$)	.0437 (β_1 2 $\rightarrow$ 1)
rocker-arm	.0083 $\pm$.0004	.0087 $\pm$.0015 (0.96 $\times$)	.0113 (counts intact)
eight (genus 2)	.0249 $\pm$.0000*	.0162$\pm$.0017 (1.5 $\times$)	.0249 (2 of 4 loops)
<i>loop group mean (H1, 3)</i>		1.52$\times$ $\pm$ 0.55	
spot	.0521 $\pm$.0081	.0237$\pm$.0000 (2.2 $\times$)	.0652 (β_2 1 $\rightarrow$ 0)
fandisk	.0538 $\pm$.0022	.0052$\pm$.0008 (10.4 $\times$)	.0571 (β_2 1 $\rightarrow$ 0)
tom-yum pot	.0221 $\pm$.0003	.0057$\pm$.0002 (3.9 $\times$)	.0223 (β_2 1 $\rightarrow$ 0)
armadillo	.0511 $\pm$.0000*	.0124$\pm$.0001 (4.1 $\times$)	.0511 (β_2 1 $\rightarrow$ 0)
horse	.0408 $\pm$.0000*	.0109$\pm$.0022 (3.8 $\times$)	.0408 (β_2 1 $\rightarrow$ 0)
<i>void group mean (H2, 5)</i>		4.87$\times$ $\pm$ 3.17	

replicates its failure modes: all five voids erased (β_2 1 $\rightarrow$ 0, every seed; the pot at 1.36 $\times$ worse Chamfer), a bob loop collapsed (β_1 2 $\rightarrow$ 1), eight’s missing pair left missing.

Observations. (1) *The H1 claim survives the family change*: bob halves loop error, zero phantom handles every seed (suppl. Fig. S8), replicating torus. (2) *The genus-2 ceiling was representational, not topological*: on the fat CGAL eight, baseline and control read only two of the four certified loops (value pinned at the unmatched bound, seed-exact); the loss *restores the missing pair* (#sig 4/3/4) and cuts value error 1.5 $\times$ at 0.89 $\times$ Chamfer — the double torus’s saturation (§5.1) was the representation’s ceiling, not genus-2’s. (3) *Sharp-featured CAD benefits most*: fandisk’s 10.4 $\times$ exceeds every analytic-family effect. (4) *Baselines can pin like controls*: armadillo and horse read the correct void count but pin its value at the same bound (.0511/.0408, seed-exact); the loss unpins both (4.1 $\times$ /3.8 $\times$) at better Chamfer. (5) *No headroom, no effect — and no harm*: rocker-arm’s one measurable loop is already baseline-correct (.0083 vs the .04–.05 band): §4’s imperfection criterion is unmet and the loss stays inert (0.96 $\times$, parity 1.00, counts intact everywhere) — the method’s honest boundary. (6) *Count-level limits persist on sub-budget organic detail*: spot’s baseline makes a phantom void in two of three seeds (#sig = 2, 1, 2); the loss clears two, retains one (1, 2, 1) — value repair reliable, count repair not guaranteed; its C1 tail pins at .0237 (sd 0), a milder floor of the double-torus kind. *And the pipeline reaches raw artist content*: the tom-yum pot (Fig. 1; suppl. Fig. S7), a raw Blender export certified rather than constructed and bundled at $M=8192$ by the same pre-registered rule (§4), cuts the void’s value error 3.9 $\times$ at Chamfer parity (0.99 $\times$), the study’s tightest seed spread (C1 .0056–.0059 vs C0 .0218–.0223); the control destroys the void (β_2 1 $\rightarrow$ 0, all seeds), pinned at the unmatched bound (.0223) — count, not value, is the verdict. Two caveats transfer: its baseline already reads the correct count (a value-accuracy win, like bob),

and ground truth is certified, not constructed. Group statistic (pre-registered): loops 1.52 $\times$ $\pm$ 0.55 (2/3 pass), voids 4.87 $\times$ $\pm$ 3.17 (5/5); the passing wave spans 1.5–10.4 $\times$ (suppl. §G).

6. Mechanism and discussion

A local signal cannot carry a global property — sparse tugs can. A pixel’s photometric gradient reaches a few vertices; topology is global and categorical — a wrong genus is not approached smoothly; the blindness probes quantify the gap (§2) — no geometric proxy stands in. The persistence term supplies the measurement with a usable gradient: each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices — *precise but sparse* pulls, iterated by re-pairing; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort along this axis: loops respond to value pressure (H1 win), wide shells to both channels (composition best), components to neither (H0 null) — and nothing responds where nothing is broken (rocker-arm’s no-headroom null, §5.2).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void’s death pins at that equilibrium (seed-identical) and drags while near-diagonal phantom bars accumulate (suppl. Fig. S1) and thin structures collapse (cube β_2 , torus β_1); the loss has no length scale — it pulls the death coordinate itself. The same gradient budget through the same channel *without* topological information is actively harmful: C1’s wins cannot be extra-regularization artifacts.

Measurability at the loss’s density. A persistence loss can only enforce features that clear the significance floor at its density M : the torus void and two of the double torus’s four loops do not, at $M=2048$. We treat this as a principle, not a nuisance: the bundle is built and audited at the same M , sub-floor features are excluded, and the double torus is evaluated against its two measurable loops — all the soup can express at feasible budgets.

The floor is quantifiable: significance means lifetime $> 6 r_{\text{med}}(M)$, and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090 $\rightarrow$.0028 over $M=2048\rightarrow 20000$; ratio 3.16 vs $\sqrt{9.77}=3.13$; suppl. Fig. S10). The double torus’s tube loops live at lifetime .045–.046 on the clean surface: at $M=2048$ that is 0.83–0.85 $\times$ the floor — invisible by a 15% margin, clearing it only from $M \approx 3 \times 10^3$ — while at the evaluation density ($M=20000$, floor .0171) they clear it 3.1 $\times$; the binding constraint is the *representation* (the soup never expresses the tubes): the *measurement* floor recedes as $M^{-1/2}$, the *representation* floor is set by

N and no M lowers it — the double torus sits on the second. The torus quantifies the noise interaction: its second loop clears the clean floor by only $1.33\times$ — a margin live-cloud noise consumes, leaving recruitment as its only gradient path (C6), the loss reaching below its own floor (suppl. §C). Suppl. Table S3 tabulates counts across M ; the pot and the group wave ran the rule *prospectively* (§4; suppl. §G), validated in-loop (Table 2); adaptive density is the natural extension.

Prescription. Correct topology in the *loss*; allocate *wide*; combine — no schedule needed once the loss is calibrated against the photometric gradient.

7. Limitations

Synthetic, single-machine. All quantitative claims are on synthetic renders with known topology: an analytic family plus eight external genus-known meshes in two groups (§5.2; one a certified raw artist mesh, the intended scan path), drawn from public repositories, not a curated benchmark suite. All surfaces are closed except a first *open-surface* probe, pre-registered and passed: two bowls cut from the sphere (openings $0.14R/0.50R$, certified disk topology) still read one significant void (the designed rim’s H1 never clears the floor); the loss cuts its error $4.2\times/4.1\times$ at better-than-baseline Chamfer (ranges disjoint); the control erases the wide bowl’s void in 2/3 seeds (suppl. §F). C7’s Gaussian probe bounds noise response, not scan realism (outliers, missing regions, registration error). Real scans remain future work: their boundaries add *noise-born* H1 bars (unlike a designed rim); bar-filtering is the near-term step.

Density floor. The loss cannot resolve features below the significance floor at its density (torus void; two of four double-torus loops at $M=2048$) — audited, but a real ceiling. Count-level repair of sub-budget detail is likewise out of scope (spot, §5.2).

Fixed target diagram. The target bundle presumes known ground-truth topology (a prior scan or template — not blind capture); template-free designs (total-persistence minimization, a target-*genus* penalty, class-level diagram priors) are untested.

Control strength. The repulsion control is aggressive (it destroys topology at calibrated magnitude); a half-radius variant is still $1.6\times$ worse (sphere .0972 vs .0623): verdicts do not hinge on control tuning.

Nondeterminism. CUDA training is not bit-reproducible (borderline prune/split decisions flip under float drift at resampling); tail CVs run 7–10% for baseline and loss arms alike, a loss-identical re-run pair (sphere C6 $\equiv$ C1) landed 6% apart — pure noise; statements follow §4’s reporting policy.

Recruitment guarantees. The zero-gradient pathology it repairs is provable (Lemma 1); its own self-correction premise is empirical (suppl. §C) — pathological geometries could defeat it; a failure-case study is future work.

8. Conclusion

We moved topology from the allocation channel into the objective: a matched persistence loss with a Gabriel-*certified* backward and a load-bearing recruitment term, every gain beating a norm-matched control at Chamfer parity. The hypotheses hold (RH1 objective root cause; RH2 topology-specific; RH3 channels compose); verdicts replicate across two external groups and degrade gracefully under noise. Prescription: topology in the *loss*, allocate *wide*, combine; next: real scans. **Reproducibility:** code, scenes, bundles, logs and seeds ship on acceptance (nondeterminism: §7).

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Supplementary Material

Appendices A–H. The main text refers to these sections as “suppl. §A” – “suppl. §H” and to their floats as “Table S1”, “Fig. S1”, and so on.

A. The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer *equal by construction* (40k samples, metrics normalized by the ground-truth bounding diagonal; Table S1). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff₉₅ actually *prefers* the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

(2) The prior arms, and why voids are “width-primary”. The study biases DiffSoup’s own resampler with a precomputed importance field built from the target’s significant features. Each significant feature f (lifetime p_f above the significance threshold) contributes Gaussian kernels at the coordinates C_f of its birth/death-simplex vertices — the loop-filling triangle, void-filling tetrahedron, or component-merging edge — sized by the feature’s own death scale:

$$\phi_d(q) = \sum_{f: \dim f=d} \sum_{c \in C_f} w_f e^{-\|q-c\|^2/(2\sigma_f^2)}, \quad w_f = p_f/s^2, \\ \sigma_f = s \text{ clip}(\sqrt{\text{death}_f}, 3r_{\text{med}}, \tfrac{1}{2}\text{diag}), \quad (5)$$

evaluated on a dense target-surface sample, normalized per dimension to $[0, 1]$ (99.5th percentile), combined by max; a query takes its nearest surface sample’s value. The spread knob s widens each kernel while preserving its injected surface mass ($\sigma \rightarrow s\sigma$, $w \rightarrow w/s^2$, so $w\sigma^2$ is s -invariant): $s=1$ is the concentrated prior (B2), $s=3$ the spread prior (B4) — the field C5 reuses. The field enters resampling at strict budget parity through two levers: prune protection — faces drop in order of $\text{vis} + \lambda Q_{0.9}(\text{vis}) \cdot \text{imp}$, removing exactly the required count — and respawn bias — top-importance visible faces subdivide first, then the protective order re-prunes to the cap. Remaining arms: B0 baseline; B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. On the void class (Table S2; $N=1200$, five paired

seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ , sphere 2.2σ , cube tie); one-shot placement washes out entirely ($B1 \approx B0$, .0546 vs .0535 on the sphere). Hence the verdict quoted in the main paper’s introduction: the prior’s topological value is carried largely by its spatial *width*.

(3) No prior shape repairs loops. On the torus ($N=700$, five paired seeds, true $\beta_1=2$) every arm fails the topology-specificity test: the best condition is the *non-topological* wide control B5 (.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) *and* manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of the main paper, and the class the loss channel wins there (main paper §5.1, Table 1).

B. Diagnostics: density sensitivity and diagram trajectories

Sampling density vs. measurability. Table S3 counts the significant bars of the clean target surfaces across sampling densities under the self-calibrating floor $6r_{\text{med}}(M)$ of the main paper’s §6. It confirms the three density claims made in the paper: the sphere’s void clears the floor at every tested density; the torus needs $M=2048$ before its *second* loop is measurable, and its own void becomes measurable only at $M=4096$ (hence the H1 restriction at $M=2048$); and the double torus’s tube loops cross the floor between $M=2048$ and $M=4096$ — consistent with the $M \gtrsim 3 \times 10^3$ estimate there — so all four loops are measurable at $M=4096$.

Diagram trajectories: the pinning mechanism, visualized. Fig. S1 shows the live H2 diagrams of one sphere seed under C0/C1/C2 at four training steps. All three arms start with the void essentially on target (initialization samples the SfM point cloud), and photometric training alone drags it away (the baseline settles $\approx .06$ below). The loss arm dips identically while the ramp is inactive, then pulls the void back to within $\approx .016$ of target — matching its measured tail. The control arm is qualitatively different: the void’s death is dragged progressively down ($\approx .23 \rightarrow .15 \rightarrow .13$) while a carpet of phantom mid-persistence bars

Table S1. Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Cham.% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

Table S2. Allocation study, void class (H2), $N=1200$, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

	B0	B2 conc.	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	.0357	.0374
cube	.0579	.0409	.0403	.0273	.0270
cylinder	.0545	.0440	.0477	.0360	.0402

Table S3. Significant bars (H1 / H2) on the clean target surfaces vs. sampling density M ; floor = $6 r_{\text{med}}(M)$, seed 0.

	$M=512$	$M=1024$	$M=2048$	$M=4096$
sphere	0 / 1	0 / 1	0 / 1	0 / 1
torus	1 / 0	1 / 0	2 / 0	2 / 1
double torus	0 / 0	2 / 0	2 / 0	4 / 0

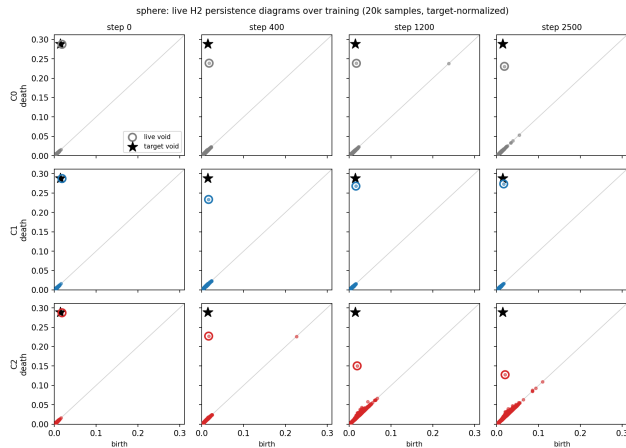


Figure S1. Live H2 persistence diagrams over training (one sphere seed; 20k samples, target-normalized; star = target void bar, circle = most persistent live bar). *Top*, baseline: the void drifts off target. *Middle*, loss: the void returns to the target once the ramp activates. *Bottom*, repulsion control: the void is pinned and dragged away while near-diagonal phantom bars accumulate.

accumulates near the diagonal — the repulsion-equilibrium signature discussed in the main paper’s §6.

The genus-2 stretch case, in full. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over five seeds; raised from two in revision, identical values throughout), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study’s floor, §A) and dominate both diagram metrics regardless of guidance. The loss adds no phantoms ($\# \text{sig } \beta_1 = 2$ every run) at marginally best Chamfer.

Sensor-noise internals (C7/C7h). Recruitment absorbs the injected noise: at $\sigma=1\%$ the torus’s second loop is sub-threshold at every refresh, with zero unreached targets. Calibration is the noisiest stage (λ_{peak} spreads $\approx 3\times$ across seeds), yet tails stay tight — consistent with the flat ρ response reported in the main paper.

The tom-yum pot: challenge inventory and prospective floor protocol. The generality wave’s artist mesh (main paper §4, §5.2) compounds the failure modes real scans will bring. Table S4 inventories them with the measured consequence of each — every row is backed by the ingest certificate or by a measurement in this study, none is asserted. Table S5 records the staircase behind the flue row: the certified shell’s significant-bar signature across sampling density, which fixed the observable class (H2) and the bundle density ($M=8192$) *before* any training run.

C. Additional result figures and the repair gate

Fig. S2 shows the generality-wave training trajectories (main paper §5.2, Table 2); Fig. S3 the per-seed tail spread behind the sphere row of main-paper Table 1; Fig. S4 the torus significant-count check behind §5.1’s zero-phantom-handles claim.

The stage-3a gate (main paper §5). Fig. S5 shows three of the four repair probes run before integration: the loss alone (no photometric term, Adam directly on point positions) must repair defective clouds. The predicted location-blind pathology of recruitment did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises

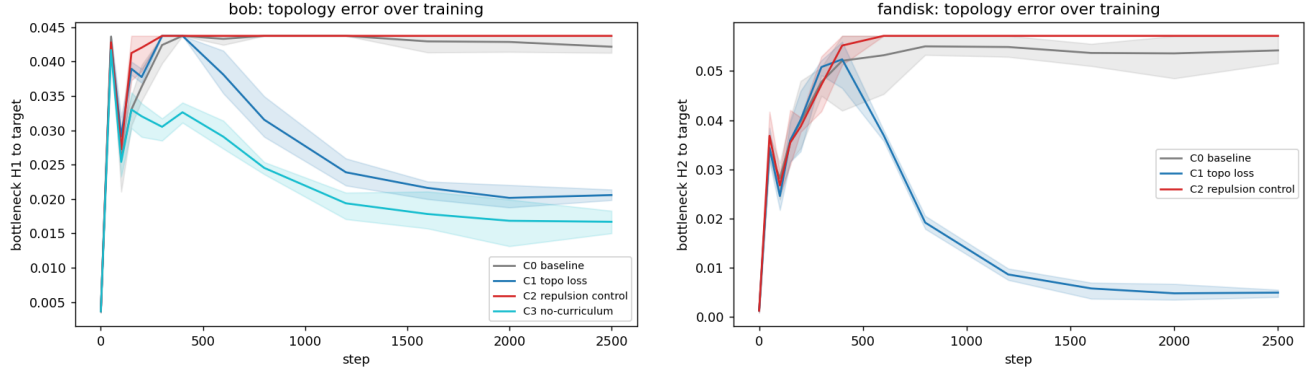


Figure S2. Generality trajectories (line = seed mean, band = seed min–max). *Left*: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with $\# \text{sig H1} = 2$ in every seed, while the control rides the baseline after collapsing a loop. *Right*: fandisk (CAD) — the largest effect measured in this study ($10.4\times$).

Table S4. Tom-yum pot challenge inventory: what makes the shape hard, and the measured consequence in this study.

challenge	measured consequence
raw artist export (6,318 vertices): 9 open shells, mixed quad/12-gon faces, 50 boundary edges after exact weld, 232 non-manifold junction edges	no trustworthy ground-truth topology as exported; ingest-and-certify required (weld, offset-solidify into a thin shell via an exact unsigned distance field + marching cubes at iso ε , drop 804 enclosed pocket shells)
certified shell: one genus 3, $\beta=(1, 6, 1)$ exact, 190,990 vertices	ground truth known by <i>certificate</i> (exact simplicial homology; watertightness cross-check; bit-identical rebuilds), not by construction
six thin handle/vent loops	every H1 loop sits below the significance floor at every working density — H1 is unusable as the loss observable
narrow flue mouth, capping at small α	the flue interior reads as an enclosed on-axis chamber — an H2 void, first measurable at $M=8192$ (margin $1.20\text{--}1.23\times$, five sampling seeds); the floor rule picks class and density prospectively (Table S5)
in-loop outcome	C1 cuts the void's error $3.9\times$ at Chamfer parity; the repulsion control <i>destroys</i> it ($\beta_2 1\rightarrow 0$, all seeds, $1.36\times$ worse Chamfer) — main paper Table 2

the recruited bar's death monotonically toward target — the runaway concentrates on the only profitable direction.

D. A decade of reconstruction, cut by optimization philosophy

The main paper's related work argues one sentence that this appendix substantiates in breadth: reconstruction methods of the differentiable decade, whatever their representation, optimize *geometric* objectives (photometric fidelity plus ge-

Table S5. Tom-yum pot staircase (seed 0; the $M=8192$ margin spans seeds 0–4): significant-bar signature $(\beta_0, \beta_1, \beta_2)$ of the certified shell vs. sampling density M , floor = $6r_{\text{med}}(M)$. The bundle is built at the first density where the discriminating feature clears ($M=8192$: exactly one significant bar, H2, stable to re-sampling at $\sim 10^{-5}$).

	$M=2048$	$M=4096$	$M=8192$	$M=20000$
sig. bars	(1, 0, 0)	(1, 0, 0)	(1, 0, 1)	(1, 0, 1)

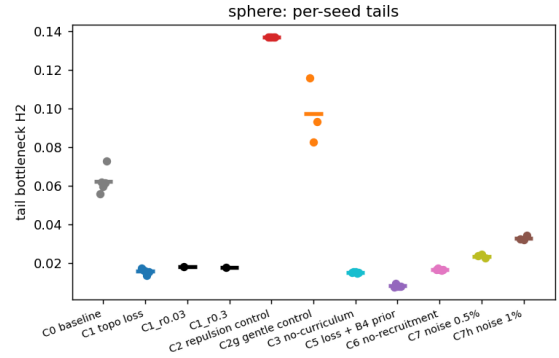


Figure S3. Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from *above*; the ρ ablation arms are indistinguishable from $\rho=0.1$.

ometric regularizers) or steer *sampling* under such an objective — topology, when correct, is correct by construction, by extraction, or by accident, never because the training objective measured it. Table S6 tabulates representative methods under this cut, including the nearest concurrent topology-aware entries; the subsections survey each family and the adjacent literatures this paper draws machinery or discipline from.

Table S6. Representative reconstruction methods by optimization philosophy. “Topology in objective?” asks whether the *training objective* measures a topological quantity of the reconstructed *surface* (not whether the representation constrains topology, and not image-space summaries). Bottom block: the nearest topology-aware concurrent work (§D.7).

method	repr.	philosophy	topology in objective?	topology from
NeRF [69]	density field	geometry	no	emergent
OccNet [68]	occupancy	geometry	no	construction (closed iso)
DeepSDF [76]	SDF	geometry	no	construction (closed iso)
NeuS [94]	SDF	geometry	no	construction (closed iso)
Neuralangelo [63]	SDF	geometry	no	construction (closed iso)
Pixel2Mesh [93]	template mesh	geometry	no	frozen at template genus
Point2Mesh [38]	template mesh	geometry	no	frozen at template genus
Mesh R-CNN [32]	voxels → mesh	geometry	no	fixed by voxelization
DMTet [86]	grid + marching	geometry	no	extraction over the grid
nvdiffrec [72]	grid + marching	geometry	no	extraction over the grid
FlexiCubes [87]	grid + marching	geometry	no	extraction over the grid
Shape-as-Points [78]	points → implicit	geometry	no	construction (closed iso)
AtlasNet [34]	patch soup	geometry	no	unmeasured
DSS [101]	point soup	geometry	no	unmeasured
3DGS [58]	Gaussian soup	geometry	no	unmeasured
2DGS [47]	surfel soup	geometry	no	unmeasured
SuGaR [35]	Gaussians → mesh	geometry	no	post-hoc extraction
MiLo [36]	Gaussians+mesh	geometry	no	in-loop extraction, unmeasured
Triangle Splatting [40]	triangle soup	geometry	no	unmeasured
Radiant Tri. Soup [11]	triangle soup	geometry	no	unmeasured (proximity forces)
Mini-Splatting [26]	Gaussian soup	sampling	no	unmeasured (budgeted resample)
DiffSoup [91]	triangle soup	geometry	no	unmeasured
DiffSoup + prior (§A)	triangle soup	sampling	no (steers only)	unmeasured; fails on loops
Topology-GS [88]	Gaussian soup	topology (image)	rendered images only	surface unmeasured
Gao et al. [29, 30]	watertight mesh	geometry	no (topology a prior)	template genus / camera prior
STITCH [51]	implicit (points)	topology	yes (β_0 , live surface)	construction + constraint
this work	triangle soup	topology	yes (live-surface persistence)	measured, optimized, budgeted

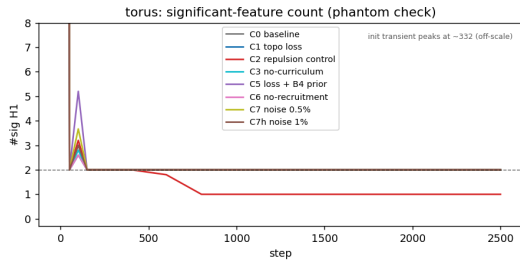


Figure S4. Torus phantom check (main paper §5.1): significant-H1 count over training (seed means; dashed line: true $\beta_1=2$). After the brief resampling transient near step 100, every loss arm sits exactly on two significant loops — zero phantom handles — while the repulsion control (C2) collapses a loop by step ~ 800 and never recovers it.

D.1. Geometry-oriented: implicit fields

Occupancy and signed-distance learning deliver closed surfaces by construction [68, 76, 77], and a fitting stack matured around the level set: Eikonal regularization [33], surface rendering [75, 98], SDF-based volume rendering [94, 99], geometric priors [102], high-resolution and accelerated variants [63, 71, 95], meshing for deployment [100], and

kernel-based point-cloud fitting [49]. Across all of them the training objective is photometric or geometric; the guarantee of consistent topology comes from the representation, its *correctness* (the right genus for the scene) is emergent, and the price is dense field evaluation plus post-hoc extraction — the cost regime the soup family exists to avoid.

D.2. Geometry-oriented: extraction, grids, and templates

Where the surface is explicit, its topology is decided by an extraction algorithm or frozen by a template. Marching Cubes and Dual Contouring fix per-cell connectivity by convention [53, 66]; Poisson reconstruction extracts an indicator iso-surface, watertight but with emergent genus [55, 56]; differentiable iso-surfacing makes extraction trainable without making topology a target [28, 72, 80, 86, 87]; learned tessellations move the per-cell decision into a network [18, 19]; and template or voxel-staged pipelines fix topology before refinement [32, 38, 93]. Robust meshing gives the same guarantee-by-construction for simulation inputs [46]. In every case topology is an input or an artifact of the algorithm — measured nowhere, optimized never.

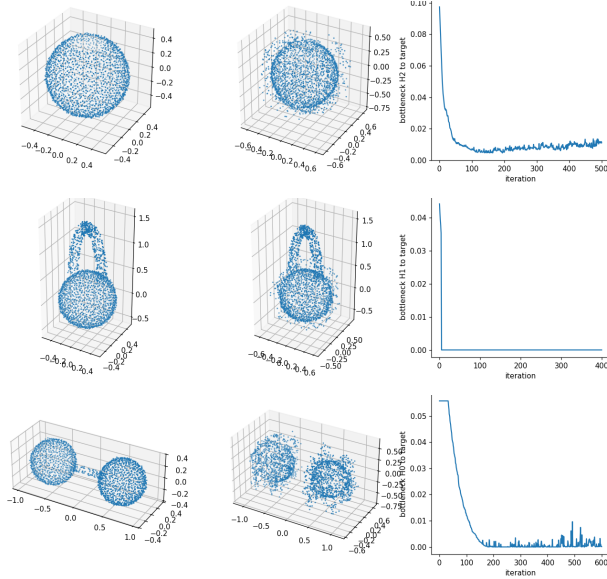


Figure S5. The stage-3a gate: the loss alone (Adam on raw points, no photometric term) repairs defective clouds; each row shows initial cloud, final cloud, and bottleneck-to-target trajectory. *Top* (T1): punctured sphere — the matched birth term closes the void ($8.8\times$). *Middle* (T2): spurious handle — the diagonal term kills the extra H1 bar. *Bottom* (T4): bridged merge with no significant live H0 bar — recruitment severs the bridge ($450\times$, $\beta_0 1\rightarrow 2$), the case where plain optimal matching provably has zero gradient. (T3, mis-spaced components, $1229\times$: omitted for space.)

D.3. Geometry-oriented: point splatting and Gaussian soups

The connectivity-free line runs from EWA surface splatting [107] through neural point rendering [2, 82, 97] to the Gaussian-splatting era [58, 103] — Pulsar’s abstract advertises the family’s freedom from mesh “topology problems” explicitly [61]. Its surface-minded members regularize geometry ever harder — flattened surfels [22, 47], planar compression [16], opacity-field extraction [104], alignment-driven meshing [35] — and MILO now extracts a mesh differentially at *every* training iteration [36]; triangle soups follow the same pattern [11, 40, 91]. Yet even with connectivity in the loop, no member of the family measures a topological quantity of what it builds: the objectives remain photometric fidelity plus geometric regularity.

D.4. Sampling-oriented: steering where a fixed objective spends

The second occupied family changes *where* primitives go, not what the objective asks of them: split-and-prune densification in 3DGS [58], budget-constrained Gaussian resampling [26], importance-driven respawn in DiffSoup [91] — and the topology-inspired allocation priors of our own study (§A), which put the family’s implicit hypothesis to a

controlled test and returned the null result the main paper builds on. Steering allocation is a real lever for *geometric* coverage; the study shows it is structurally the wrong lever for one-dimensional topology.

D.5. Topology as machinery: persistence in and around learning

The ingredients for topology-aware objectives have been maturing for two decades: persistence computation at scale [6, 67, 106], practical diagram metrics [57], stability theory [21], and vectorized representations [1, 10] on the computational side; trainable diagram layers [12, 41], filtration learning [42], and general differentiability frameworks [13, 62, 74] on the gradient side (surveyed in [14, 15]). In-loop topological losses already win in adjacent domains — representation learning [70], tubular and topologically faithful segmentation [20, 45, 89, 90], microscopy volume prediction [92], point-cloud repair and shape optimization [8, 79] — and differentiable non-persistence signals exist too [81]. What had not happened is the transfer of this machinery into image-based surface reconstruction.

D.6. Topology outside the loop: repair and benchmarks

Before differentiable pipelines, topology was corrected after the fact: spurious-handle removal [37, 96], general mesh repair [4, 52, 85] (surveyed in [5]), and soup-to-manifold conversion [46, 48]; Thingi10K quantifies how pervasive soup-like defects are in the wild [105] — our tom-yum pot’s raw export is a typical specimen. The evaluation culture matches: standard pipelines and benchmarks — COLMAP [83, 84], PMVS [27], DTU [50], Tanks and Temples [59] — score accuracy and completeness by point distances; none carries a topological-correctness metric, the blindness our Table S1 quantifies on controlled probes.

D.7. The empty cell, and its nearest neighbours

Concurrent work has begun to press on the gap from several directions, and the distinctions matter. Topology-GS [88] puts a persistence term *inside* Gaussian-splatting training — but on barcodes of *rendered images*, a topological-perceptual loss; the surface’s diagram is never measured, and densification adds primitives rather than budgeting them. The high-genus inverse-rendering line [29, 30] reconstructs meshes whose genus is guaranteed by a Gauss–Bonnet-matched template or guides camera placement with homology computed once on a reference shape — topology as a *prior*, assumed rather than measured, with ground-truth genus required up front. STITCH [51] does measure a differentiable persistence quantity on a live implicit surface — enforcing a single connected component — but from point-cloud input: no images, no primitives, no budget. Topology also enters generation as conditioning features [44], again

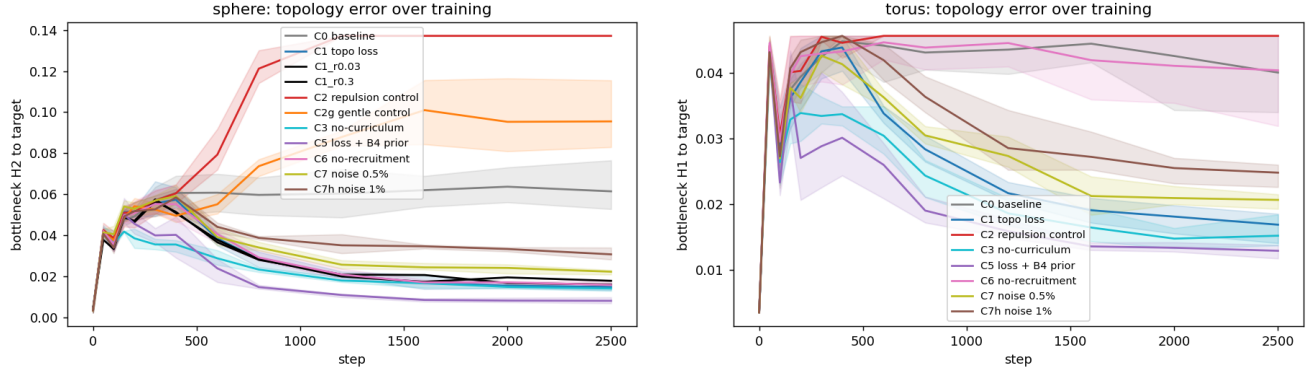


Figure S6. Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min–max). *Left*, sphere (H2): every loss arm (C1, C3, C5, C6, both ρ variants, noise pair C7/C7h) converges to a fraction of baseline; the norm-matched repulsion control C2 is *destructive*, and its gentle variant C2g still ends worse than baseline. *Right*, torus (H1), the class no prior won: same ordering, C5 best; noise arms descend shallower but never approach baseline; the no-recruitment ablation C6 rides it.



Figure S7. The *tom-yum pot*: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, $\beta=(1,6,1)$ exact (main paper §4). *Right*: cutaway — the narrow flue mouth caps at small α , so the flue interior reads as an enclosed chamber: the on-axis H2 void the floor rule selects at $M=8192$ (all H1 loops sit below the floor).

outside the reconstruction objective. Partition the three requirements — (i) a topological quantity of the evolving *surface* measured differentially in the objective, (ii) image-based reconstruction through a renderer, (iii) a fixed primitive budget — and each neighbour holds at most one; none holds all three. That cell — the bottom row of Table S6 — is what the main paper fills.

E. Per-seed values and calibration sweeps

The main paper’s reporting policy (§4) quotes Welch σ separations only where both arms ran five seeds; every three-seed verdict instead rests on disjoint per-seed ranges. Table S7 lists the underlying per-seed tail bottlenecks for the generality wave and the three-seed analytic shapes. In every row except rocker-arm’s (the no-headroom null, main §5.2) the largest C1 seed is well below the smallest C0 seed; identical values across seeds — C2’s, and the group wave’s pinned C0s — are genuine (the unmatched-target bound,

main paper §§5.2, 6; suppl. §G), not rounding.

Table S8 tabulates the two calibration-knob sweeps whose verdicts the main paper states in §3.4 and §5.1: the ρ response is flat over a $10\times$ range, and the ramp window is immaterial — all alternatives land within seed noise of the default.

F. The open-surface probe

Two open bowls are cut from the analytic sphere’s ground-truth mesh by removing a polar cap (measured openings $0.14 R$ and $0.50 R$), then certified by measurement as disk topology: one shell, exactly one boundary loop, edge-manifold, consistently oriented, Euler characteristic 1. The pre-registered bundle rule (the pot’s) applies unchanged: the density staircase decides the observable and density — and it overruled our own expectation. We predicted the wide bowl would lose its enclosed void to the rim; instead *both* bowls read exactly one significant H2 bar at every tested density (the mouth caps at small α , so the interior reads as a chamber — the pot’s mechanism on an open surface), while the *designed* rim’s H1 bar never clears the floor (Table S9). The rule therefore sets H2 at $M=2048$ for both, budget $N=1200$, loss restricted to H2 (the torus precedent for sub-floor features). A designed rim is a *real* feature of the open target, distinct from the noise-born boundary bars real scans will add — the bar-filtering step named in main §7 concerns the latter.

The C-matrix (C0/C1/C2, three seeds, protocol otherwise unchanged) passes the verdict rule on both bowls at better-than-baseline Chamfer (Table S10): the loss cuts the void’s error $4.2\times$ (narrow) and $4.1\times$ (wide) with disjoint per-seed ranges, inside the study’s $1.5\text{--}10.4\times$ span. The control replicates its closed-surface failure modes: on the narrow bowl it pins the void at its repulsion equilibrium

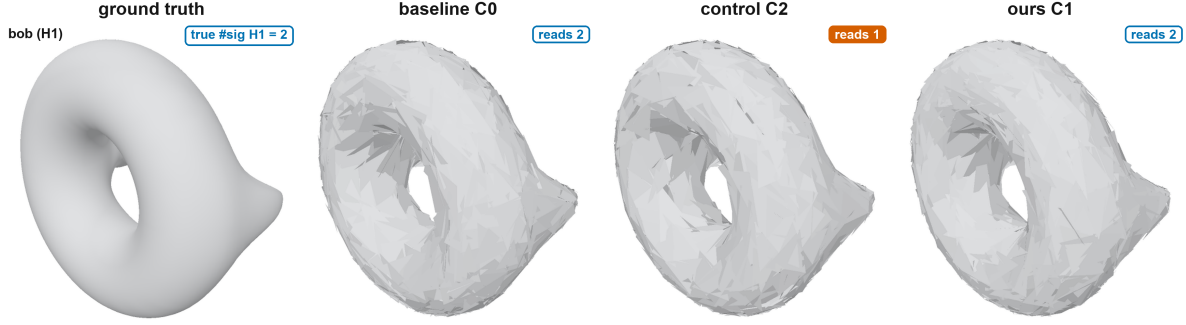


Figure S8. The loop-class row of the main paper’s Fig. 1 matrix: bob (genus 1, H1). The repulsion control’s reconstruction reads one significant loop against the true two ($\beta_1 2 \rightarrow 1$, all seeds) at $2\times$ worse Chamfer, while baseline and loss both read two — the loss at half the baseline’s tail error (main Table 2).

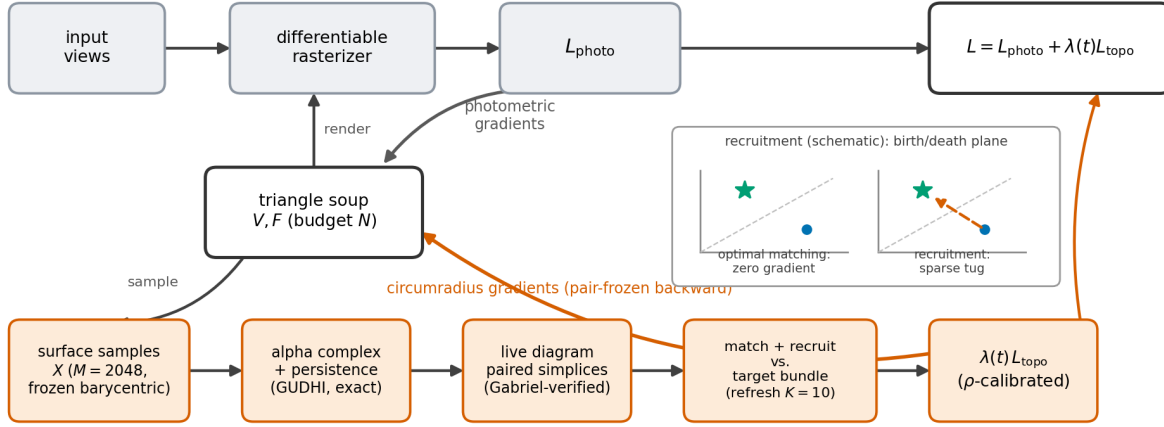


Figure S9. The system, schematically. DiffSoup’s photometric path (gray) is untouched; the topological branch (orange) samples the live soup ($M=2048$, frozen barycentric weights), computes exact alpha-complex persistence, matches and recruits against the fixed target bundle (refresh every $K=10$ steps and after every resample), and back-propagates through the pair-frozen circumradius re-expressions into the same vertices (main §3). *Inset* (schematic): an absent target bar (star) receives no gradient under optimal matching (main Lemma 1); recruitment attaches the nearest unclaimed live bar — the sparse tug.

($2.1\times$ worse than baseline, identical across seeds) at $3.5\times$ worse Chamfer; on the wide bowl it *erases* the void in two of three seeds ($\beta_2 1 \rightarrow 0$) at $3.1\times$ worse Chamfer. Every C0/C1 seed reads the correct count.

G. The group wave: expanding the external set to two groups

The generality wave of main §5.2 grew from four external meshes to eight, organized as the study’s two observable classes — the *loop group* (H1: bob, rocker-arm, eight) and the *void group* (H2: spot, fandisk, the tom-yum pot, armadillo, horse) — under a pre-registered plan (GROUP_WAVE_PLAN.md, shipped with the release bundle) that fixed the candidate pool, the selection rule, the

run protocol, and the group statistic before any training run. Per-seed tails for the new shapes join suppl. Table S7; this section records where the shapes came from and how they were selected, and Fig. S11 renders all four under the front-page convention.

Pool, certificates, honest exclusions (Table S12). Candidates were drawn from a standard test-model collection (alecjacobson/common-3d-test-models, pinned commit) plus the CGAL repository’s genus-2 *eight*; the release bundle ships fetch scripts with pinned commits and the certificates, not the meshes — redistribution terms remain the original repositories’. Every candidate passed or failed the pot’s mesh certificate (main §4: exact simplicial homology cross-checked against per-component

Table S7. Per-seed tail bottlenecks (seeds 0/1/2, run order) behind the three-seed verdicts. Spot’s C1 value repeats across seeds at this precision; the pot rows are the quicklook values behind main Table 2.

shape (dim)	C0	C1 loss	C2 control
spot (H2)	.0429 / .0584 / .0550	.0237 / .0237 / .0237	.0652 × 3
bob (H1)	.0429 / .0413 / .0437	.0200 / .0207 / .0216	.0437 × 3
fandisk (H2)	.0540 / .0515 / .0559	.0058 / .0055 / .0042	.0571 × 3
tom-yum pot (H2)	.0218 / .0223 / .0223	.0059 / .0056 / .0056	.0223 × 3
rocker-arm (H1)	.0088 / .0082 / .0079	.0090 / .0070 / .0100	.0112 / .0112 / .0115
eight (H1)	.0249 × 3	.0169 / .0175 / .0142	.0249 × 3
armadillo (H2)	.0511 × 3	.0124 / .0125 / .0123	.0511 × 3
horse (H2)	.0408 × 3	.0124 / .0083 / .0118	.0408 × 3
cube (H2)	.0560 / .0551 / .0636	.0071 / .0064 / .0086	.1346 × 3
two spheres (H0)	.0060 / .0074 / .0062	.0007 / .0009 / .0005	.0009 / .0009 / .0006

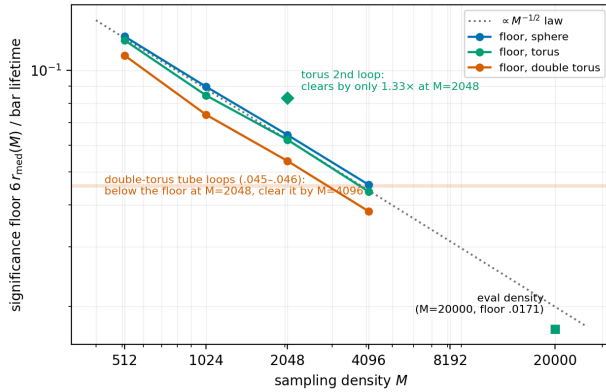


Figure S10. The measurement floor, visualized (all values from the recorded density sweep behind Table S3 and main §6). The floor $6 r_{\text{med}}(M)$ falls as $M^{-1/2}$ (dotted: the law anchored at the torus $M=2048$ point, reaching the $M=20000$ evaluation floor .0171), while a feature’s lifetime is fixed — a feature becomes measurable where the falling floor crosses it. The double torus’s tube loops (.045–.046 band) sit below the floor at $M=2048$ and clear it by $M=4096$, matching Table S3’s counts; the torus’s second loop clears by only $1.33\times$ at $M=2048$ (lifetime marked at $1.33\times$ the .0623 floor) — the thin margin that live-cloud noise consumes (main §6, C6).

Table S8. Calibration-knob sweeps (sphere, tail bottleneck). *Top*: ρ response, matched seed-0 sweep — the five-seed $\rho=0.1$ arm is main Table 1’s C1 (.0156±.0013). *Bottom*: ramp-window pilot, three seeds each.

ρ (seed 0)	0.03	0.1	0.3
tail	.0181	.0172	.0177
ramp window	10–40%	20–50% (default)	30–60%
tail	.0154±.0017	.0156±.0013	.0179±.0019

Euler characteristics, watertightness, orientation) as-is — failures were excluded, never repaired, keeping the wave’s class “watertight benchmark” (the pot remains the

Table S9. Probe staircase (seed 0): top bar lifetime over the floor, per dimension. Both bowls read exactly (0, 0, 1) significant bars at every density — one H2 void, no significant H1 — so the table gives the margins.

top bar / floor	$M=2048$	$M=4096$	$M=8192$
narrow: H2 / H1	$3.79\times / 0.47\times$	$5.50\times / 0.57\times$	$7.82\times / 0.45\times$
wide: H2 / H1	$2.34\times / 0.42\times$	$3.50\times / 0.51\times$	$4.94\times / 0.46\times$

Table S10. Open-surface probe C-matrix: tail bottleneck (mean±sd, 3 seeds; per-seed values in run order beneath); × = reduction vs C0. Both shapes **pass** the verdict rule; no Welch σ at $n=3$ (main §4).

shape	C0	C1 loss	C2 control
bowl _{narrow}	.0588±.0035	.0138±.0013 (4.2×)	.1249 (pinned)
per-seed	.0619/.0550/.0593	.0137/.0126/.0151	.1249 × 3
bowl _{wide}	.0560±.0098	.0137±.0003 (4.1×)	.0770 ($\beta_2 1 \rightarrow 0, 2/3$)
per-seed	.0448/.0602/.0630	.0141/.0136/.0135	.0770 × 3

study’s one ingest-and-solidify story). Five of eleven failed: open boundaries or non-manifold junctions in beast, cow, max-planck, ogre, and the Utah teapot.

Staircase selection, before training (Table S11). The density staircase (the pot’s pre-registered rule, unchanged) fixed each certified shape’s observable class and working density: *rocker-arm* (genus 1) enters the loop group on the torus precedent — one of its two loops clears the $M=2048$ floor (0.0929 vs 0.0613), the second (0.0204) reaches only $1.15\times$ the floor even at $M=20,000$, and the void is sub-floor at 2048, so the loss is H1-restricted; *eight* (genus 2) enters on the pot’s precedent — the full exact (1, 4, 1) is significant from $M=4096$, and the bundle is built at $M=8192$ where the fourth loop clears $1.69\times$ and the void $2.6\times$; *armadillo* and *horse* (genus 0) read the exact (1, 0, 1) at every tested density (voids $1.68\times/1.42\times$ over the 2048 floor). Two certified candidates were set aside at this stage, before any training: *cheburashka* qualifies at 2048 but grows a second significant capping H2 bar from $M=8192$ (0.0300 vs

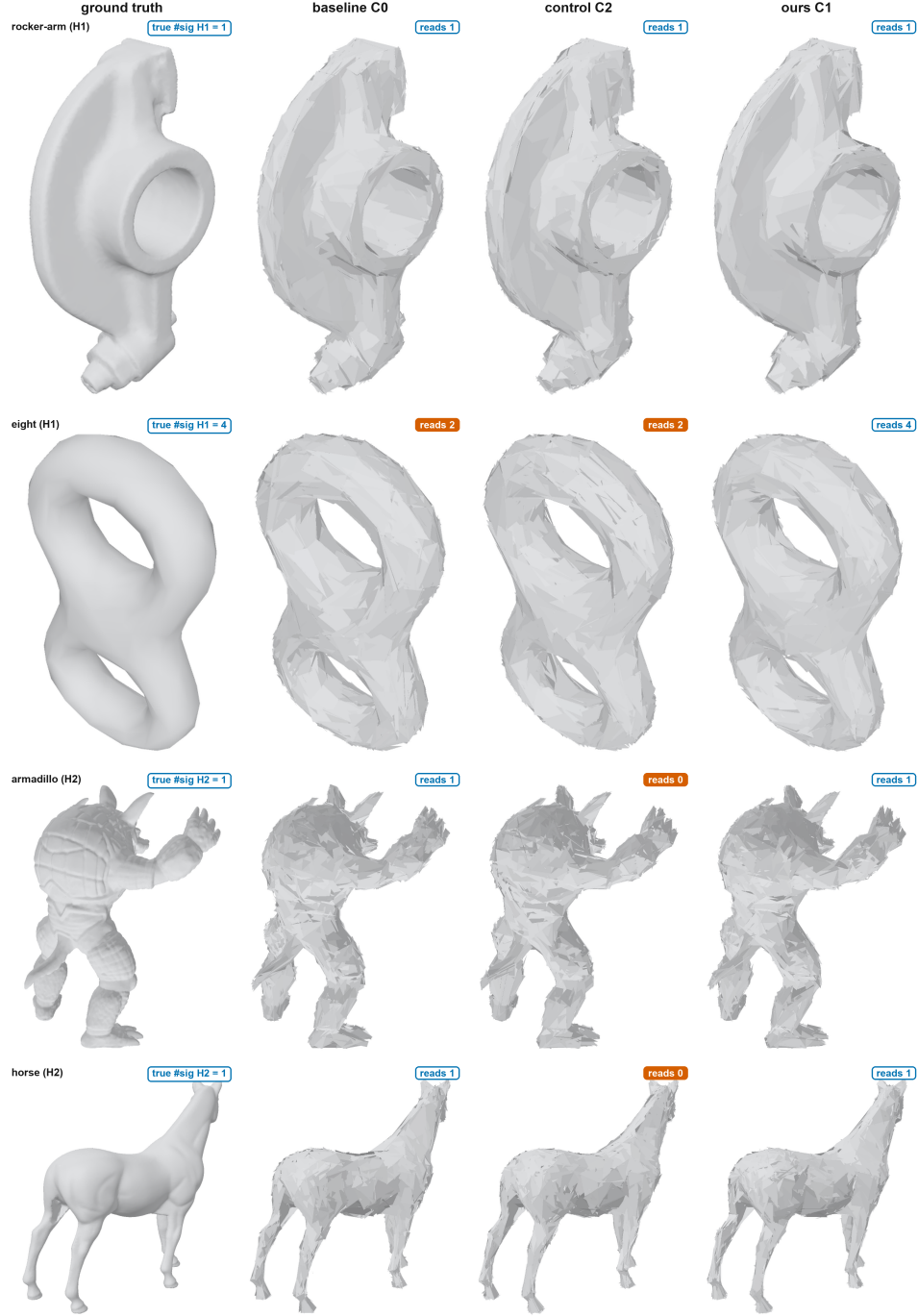


Figure S11. The group wave, rendered (ground truth / C0 / C2 / C1; seed-0 final soups; material, per-row camera and badge convention of main Fig. 1). *rocker-arm*: the no-headroom null — every arm reads the one measurable loop correctly (all badges blue; the control’s damage here is value-only). *eight*: baseline and control read only two of the four certified loops while the loss restores all four (per-seed C1 counts 4/3/4; seed 0 shown) — and the missing pair is invisible in the render, both holes being open in every cell: the deficit is a sub-floor diagram reading, exactly the blindness the paper measures. *armadillo*/*horse*: correct-count baselines whose void *value* pins at the unmatched bound (main Table 2); the control erases both voids (reads 0) with no visible geometric collapse.

floor 0.0271; 0.0312 vs 0.0171 at 20k) — a pot-flue-type concavity reading that would make the evaluation-density

phantom check ambiguous — and a denser retessellation of eight (license unstated at its source) whose staircase inde-

Table S11. Group-wave staircases (seed 0): significant Betti counts vs sampling density for the four selected shapes; bold = the pre-registered working density. rocker-arm’s second loop and 2048-density void never clear the floor (loss H1-restricted, torus precedent); eight’s full (1, 4, 1) holds from 4096 (bundle at 8192, pot precedent).

sig ($\beta_0, \beta_1, \beta_2$)	$M=2048$	4096	8192	20000
rocker-arm	(1,1,0)	(1,1,1)	(1,1,2)	(1,2,2)
eight	(1,2,1)	(1,4,1)	(1,4,1)	(1,4,1)
armadillo	(1,0,1)	(1,0,1)	(1,0,1)	(1,0,1)
horse	(1,0,1)	(1,0,1)	(1,0,1)	(1,0,1)

Table S12. Candidate disposition (selection fixed before training; plan file in the release bundle). Certificate failures are excluded, never repaired; the two staircase-stage set-asides are motivated in the text.

candidate	origin	certificate	decision
rocker-arm	INRIA / AIM@SHAPE	pass (g1)	loop group
eight	CGAL data (CC0-class)	pass (g2)	loop group
armadillo	Stanford scan rep.	pass (g0)	void group
horse	CyberWare / GaTech	pass (g0)	void group
cheburashka	I. Baran (Pinocchio)	pass (g0)	set aside
eight (dense)	Directional (lib.)	pass (g2)	set aside
beast	Autodesk	fail: non-manifold	—
cow	Viewpoint/Sun	fail: odd χ	—
max-planck	MPI	fail: 161 open edges	—
ogre	K. Ritchie	fail: 200 open edges	—
teapot	M. Newell	fail: 160 open edges	—

pently corroborates the (1, 4, 1) observable at $M=8192$.

Outcome identities behind the pins. The three seed-exact baseline values of main Table 2 are unmatched-target bounds, not coincidences: eight’s C0/C2 pin $0.0249 \approx$ half the third loop’s ground-truth lifetime at the evaluation density ($0.0498/2$), armadillo’s $0.0511 = 0.1022/2$, horse’s $0.0408 \approx 0.0817/2$ — the bottleneck cost of leaving that target bar unmatched (main §6’s pin mechanism, previously seen in the pot’s control arm).

H. Robustness of the reporting choices

The floor multiplier is not load-bearing. The significance floor $k r_{\text{med}}(M)$ uses $k=6$ throughout. Recomputing every documented bundle decision from the recorded ground-truth clouds (protocol densities, bundle seed) gives each feature’s flip point — the k at which an included feature would leave the bundle or an excluded one would enter — and *every decision in the study is unchanged for* $k \in (5.12, 6.77)$ (Table S13). The two binding features are exactly the ones the paper already flags as marginal: the double torus’s sub-floor tube loops ($0.85\times$ the $k=6$ floor) bound it below, bob’s fat-tube void ($1.13\times$) bounds it above. The verdict metric itself (bottleneck to the target diagram) contains no threshold; k only gates which bars enter bundles

Table S13. Floor-multiplier windows per shape (recorded GT clouds at the protocol bundle density): the tightest included margin and, where the protocol excludes sub-floor features, the largest excluded margin — each decision holds while k stays below/above $6\times$ that margin. Global window: $k \in (5.12, 6.77)$.

shape	tightest IN ($\times$ floor)	largest OUT ($\times$ floor)
sphere / cube	3.86 / 2.71	—
torus	1.33 (loop 2)	0.83 (void)
two spheres	2.61	—
double torus	2.23	0.85 (tube loops)
spot / fandisk	1.87 / 1.50	—
bob	1.13 (void)	—
tom-yum pot	1.22	0.46 (loops)
rocker-arm	1.52	0.74 (void)
eight	1.69 (loop 4)	—
armadillo / horse	1.68 / 1.42	—
bowls (narrow/wide)	3.79 / 2.34	0.47 / 0.42 (rims)

and counts.

The pair-frozen plan is stable across refreshes (Fig. S12). Every run since the refresh-log schema records the plan’s composition (matched / diagonal / recruited / target-unreached, per dimension) at each of its ~ 200 refreshes. Three regimes cover the study: the torus without recruitment (C6) shows the chronic gap the ablation collapses on — one matched loop and one unreached target at *every* refresh; on the genus-2 eight, recruitment claims the two sub-threshold loops from the first active refresh and *hands them over* to the matched set as training makes them significant (matched $2 \rightarrow 3$ —late, recruited $2 \rightarrow 1$ —0, unreached 0 throughout); the tom-yum pot settles to a stable one-bar matched plan within the first ~ 250 active steps. No plan oscillation beyond single-bar hand-overs is observed, and the per-refresh Gabriel-failure count is zero in all three runs, as everywhere (main §3.2).

Cost scaling, measured. The persistence step — the refresh’s dominant cost — measured on the torus cloud: 174 ms at $M=2048$ (reproducing §3.2’s recorded refresh figure), 388 ms at 4096, 740 ms at 8192, 1.92 s at 20,000 — an $11.0\times$ cost for $9.8\times$ the points, consistent with the near-linear $O(M \log M)$ scaling. GPU memory: a single-seed probe (torus, 2,500 steps, whole-GPU used memory sampled at 400 ms) peaks at 3,649 MiB for the baseline and 4,035 MiB with the loss (+386 MiB); the persistence computation itself runs on the CPU (GUDHI), off the GPU budget.

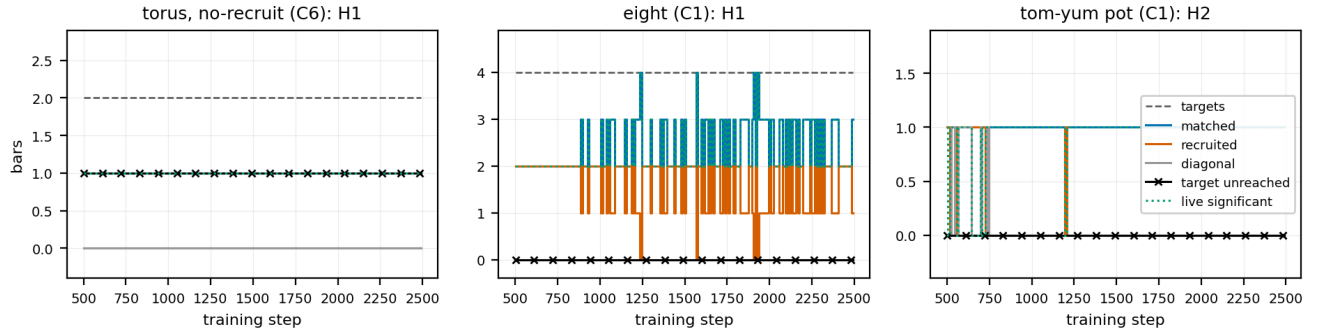


Figure S12. Per-refresh plan composition from the recorded refresh logs (seed 0; ~ 200 refreshes each; no new runs). *Left*: torus without recruitment (C6) — one loop matched, one target unreachable at every refresh: the chronic gap behind the C6 collapse (main §5.1). *Middle*: eight (C1) — recruitment claims the two sub-threshold loops early and hands them to the matched set as training makes them significant. *Right*: the tom-yum pot (C1) — a stable one-bar matched plan after the first ~ 250 active steps. Dashed gray = target count; in the right panel it coincides with the matched line.